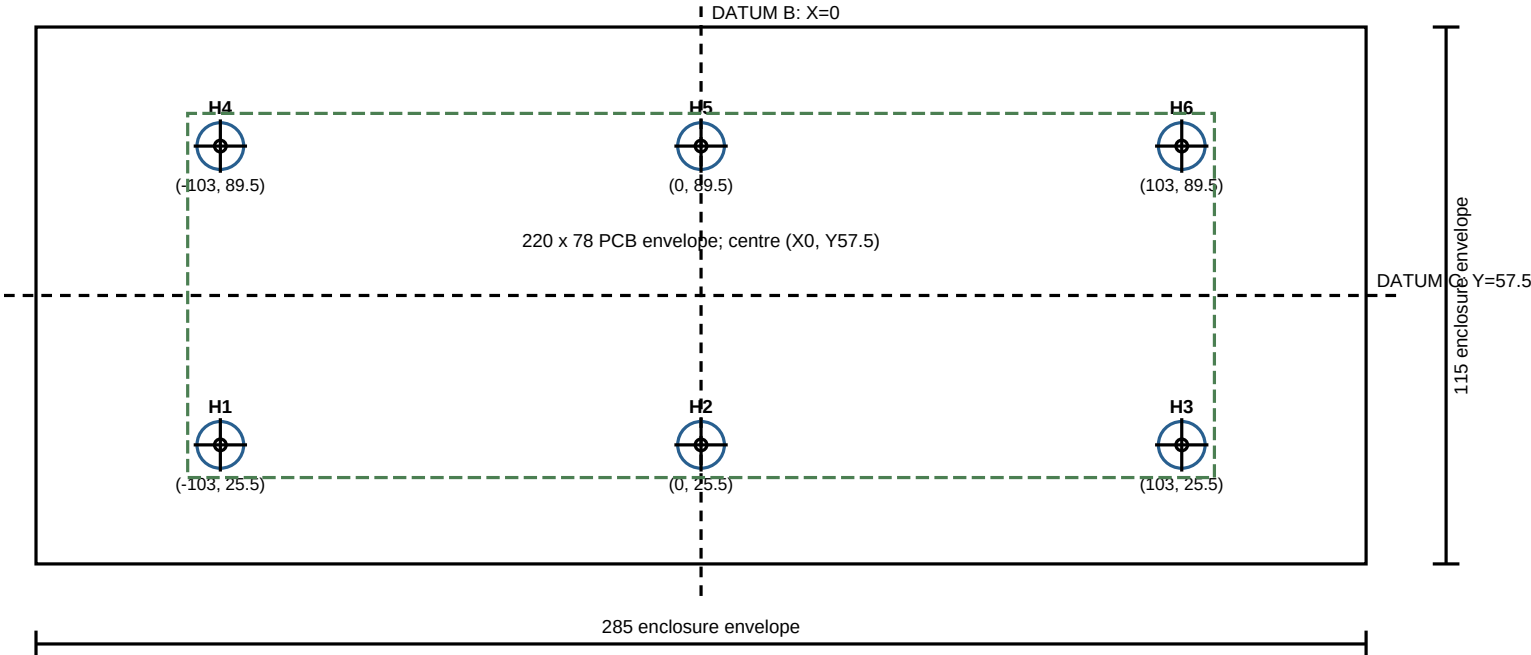


RELEASE CANDIDATE - ENGINEERING APPROVAL REQUIRED - NOT FABRICATION AUTHORITY

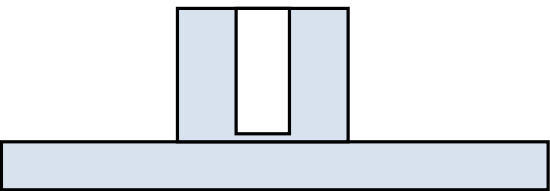
BRMC Consumer v1.0 CNC 6061 enclosure base - six threaded bosses

UNITS: mm | SCALE: noted | DO NOT SCALE

TOP VIEW - 1.75 pt/mm



SECTION A-A - THREADED BOSS



A = exterior bottom, Z0
Inner floor top Z3.00 nominal
Boss OD 10.00 +/-0.10
Height 5.00 +/-0.05
Support plane Z8.00 +/-0.05 to A
M3 x 0.5-6H blind
Full thread depth 4.30 MIN
Tap drill dia 2.50 +0.10/-0.00
Tap drill depth 5.30 +/-0.20
Residual exterior floor 2.70 nominal

BOSS / HOLE SCHEDULE - ENCLOSURE DATUM COORDINATES

REF	X	Y	TOP Z	BOSS OD	HEIGHT	THREAD	FULL THREAD	PCB HOLE
H1	-103	25.5	8.00 +/-0.05	10.00 +/-0.10	5.00 +/-0.05	M3x0.5-6H	4.30 MIN	dia 3.20 NPTH
H2	0	25.5	8.00 +/-0.05	10.00 +/-0.10	5.00 +/-0.05	M3x0.5-6H	4.30 MIN	dia 3.20 NPTH
H3	103	25.5	8.00 +/-0.05	10.00 +/-0.10	5.00 +/-0.05	M3x0.5-6H	4.30 MIN	dia 3.20 NPTH
H4	-103	89.5	8.00 +/-0.05	10.00 +/-0.10	5.00 +/-0.05	M3x0.5-6H	4.30 MIN	dia 3.20 NPTH
H5	0	89.5	8.00 +/-0.05	10.00 +/-0.10	5.00 +/-0.05	M3x0.5-6H	4.30 MIN	dia 3.20 NPTH
H6	103	89.5	8.00 +/-0.05	10.00 +/-0.10	5.00 +/-0.05	M3x0.5-6H	4.30 MIN	dia 3.20 NPTH

MANUFACTURING / CONTROL NOTES

- MATERIAL: 6061-T6 aluminum to ASTM B209/B221. PROCESS: CNC mill from billet/plate; integral bosses; no inserts.
- FINISH: Type II Class 2 black anodize, sealed, 12-25 um per MIL-PRF-8625. Mask all threads, datum/interface and electrical bond surfaces as specified by assembly drawing.
- DATUM A: exterior bottom plane Z0. DATUM B: longitudinal center plane X0. DATUM C: transverse center plane Y57.5. Boss-axis true position dia 0.20 to A|B|C.
- Six boss tops coplanar within 0.10. Boss axes perpendicular to A within 0.10 over 5 mm. Break edges 0.2-0.5; remove burrs. Surface finish Ra 3.2 um unless noted.
- FASTENER: M3x6 ISO 7380-1 A4-70 button-head screw plus 0.5 mm PA66/PEEK washer. Nominal engagement 3.9 mm; verify no bottoming. Assembly torque 0.35 +/-0.05 N.m with marine-compatible anti-seize.
- GENERAL TOLERANCES: ISO 2768-mK. Critical dimensions and GD&T shown here take precedence. STEP models tap-drill geometry; this drawing controls the M3 thread.
- Nominal OD10 x 5 boss envelopes have zero positive-volume intersections with the 19 non-base solids in the hash-controlled v0.8 assembly. See interference CSV.
- RELEASE HOLD: populated PCB/component bodies, connector mates, routed harnesses, thermal hardware and service-tool envelopes are absent; tolerance-aware final assembly review and approvals remain mandatory.
- CONTROLLED SOURCE SHA-256: 1e1f78229c2624be0a0ecc7766795e55c19090104cca487c9a2ebd9da287a810.

BRMC Consumer v1.0 CNC 6061 enclosure base

DOC: BRMC-MECH-EVT-010 | DRAWING: 010-BASE-01

STATUS: RELEASE CANDIDATE - APPROVAL HOLD

REV: B

DATE: 2026-09-03

SHEET: 1 / 1